

LANGUAGE: PYTHON PROGRAMMING
LEVEL: BASIC-INTERMEDIATE

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	Corporate Use Case
Day 1	Python Introduction & Setup	Introduction to Python, Advantages of Python, Python 2 vs Python 3, Installation, IDE Setup, Python Interpreter, print(), input(), help(), dir()	LeetCode: Fizz Buzz HackerRank: Python If-Else	Automating Excel Reports for Finance Teams
Day 2	Python Basics	Variables, Data Types, Type Conversion, Operators, Strings, Built-in Functions, Input/Output	LeetCode: Palindrome Number, Valid Anagram HackerRank: Python Strings	Employee Data Validation System
Day 3	Control Statements & Loops	if-else, Nested Conditions, for Loop, while Loop, break, continue	LeetCode: Power of Two, Fizz Buzz HackerRank: Loops	Automated Ticket Routing System
Day 4	Functions in Python	Function Definition, Parameters, Return Values, Variable Scope, Recursive Functions	LeetCode: Climbing Stairs, Two Sum HackerRank: Python Functions	Reusable Validation Module
Day 5	Collections in Python	Lists, Tuples, Dictionaries, Sets, List Comprehensions, enumerate(), zip()	LeetCode: Contains Duplicate, Majority Element HackerRank: Lists, Dictionaries and Maps	Customer Order Processing Dashboard
Day 6	Object-Oriented Programming (OOP)	Classes, Objects, Constructors, Inheritance, Method Overriding, Encapsulation, Polymorphism	LeetCode: Design Parking System HackerRank: Classes and Objects	Employee Management System
Day 7	Exception Handling & Debugging	try-except-finally, Custom Exceptions, Assertions, Logging Basics, PEP8 Coding Standards	LeetCode: Missing Number, Single Number HackerRank: Exceptions	Payroll Error Handling System
Day 8	File Handling & Modules	Reading/Writing Files, CSV Handling, JSON Basics, Modules, Packages, import Statement	LeetCode: Matrix Diagonal Sum, Rotate Array HackerRank: File I/O	Security Log Analyzer
Day 9	Database Connectivity	sqlite3 Basics, CRUD Operations, Database Transactions, SQL Queries using Python	LeetCode: Move Zeroes, Majority Element HackerRank: SQLite Basics	Employee Attendance System
Day 10	NumPy & Pandas Basics	NumPy Arrays, Pandas DataFrames, Filtering, Sorting, Data Cleaning	LeetCode: Maximum Subarray HackerRank: Numpy Arrays	Sales Analytics Dashboard
Day 11	Data Visualization & ML Basics	Matplotlib Basics, Charts, Introduction to Data Science, Basics of Scikit-Learn	LeetCode: Binary Search HackerRank: Statistics Basics	Customer Feedback Analysis Dashboard
Day 12	Advanced Python Concepts	Lambda Functions, map/filter/reduce, Decorators, Generators, Multithreading Basics, REST API Basics	LeetCode: Top K Frequent Elements, Longest Substring Without Repeating	Enterprise Automation Pipeline

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			Characters HackerRank: Collections, Itertools	
Day 13	Coding Practice + RDBMS Concepts	First Half – Coding Practice: Mock Coding Interviews, Collections-Based Problems, Debugging. Second Half – RDBMS: DBMS vs RDBMS, Keys, Normalization, SQL Basics, Joins	LeetCode: Valid Parentheses, Contains Duplicate HackerRank: SQL Basic Select, Dynamic Array	Enterprise Reporting System
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Cloud Computing Basics, Git & GitHub Workflow, Docker Basics, Kubernetes Basics, CI/CD Pipelines, GitHub Actions/Jenkins Basics	LeetCode: Single Number, Move Zeroes HackerRank: Linux Shell Basics, Git Tutorials	Python Application Deployment using Docker & CI/CD

Important Python Libraries

Library	Usage
NumPy	Numerical Operations
Pandas	Data Analysis
Matplotlib	Data Visualization
Scikit-Learn	Machine Learning Basics
sqlite3	Database Connectivity
json	JSON Processing
threading	Multithreading

Recommended Platforms

LeetCode

HackerRank Python

GeeksforGeeks Python

Be Agile..Be in Demand